Hunting Exoplanets with Machine Learning

Classifying NASA Kepler Objects of Interest — FALSE POSITIVE · CANDIDATE · CONFIRMED

India High School Exoplanet Data Challenge (Celesta × NASA) | Dataset: NASA Exoplanet Archive — Kepler KOI Cumulative Table (DOI 10.26133/NEA4) | Repo: github.com/matrixmavericks/exoplanet-koi-classifier

HEADLINE RESULT — Tuned XGBoost, 3-class

Cross-validated macro-F1 = **0.928 ± 0.007** (5-fold) • Accuracy = **0.947** • FALSE POSITIVE recall ≈ **99%** • Binary (real vs. false) macro-F1 = **0.993**

1 The Problem

NASA's Kepler telescope measured the brightness of about 200,000 stars, searching for the tiny, repeating dips in light — called **transits** — that occur when a planet passes in front of its star. But most dips are impostors: eclipsing binary stars, instrument noise, or background blends. Each of the 9,564 Kepler Objects of Interest (KOIs) in this dataset has been sorted by experts into **CONFIRMED** (a validated planet), **CANDIDATE** (a real-looking signal not yet confirmed), or **FALSE POSITIVE** (an artifact). My goal was to reproduce that expert triage automatically from 140 columns of transit geometry and stellar measurements — the same problem NASA scientists face when vetting the Kepler pipeline.

2 Exploratory Data Analysis & Data Cleaning

The three classes are imbalanced — false positives make up roughly 51% — so I evaluate models by **macro-F1** (the unweighted mean of per-class F1) rather than accuracy, which stops a lazy 'always guess false positive' model from looking good. Exploring the data revealed 19 columns that are entirely empty (dropped), and a great deal of missing data that is *not* random: only CONFIRMED planets are ever given a name, for example. For genuine numeric gaps I use median imputation (robust to the heavy tails common in astronomy), and the gradient-boosted trees also handle missing values natively. Distribution plots confirmed usable physics: confirmed planets sit at high signal-to-noise with sensible radii, while false positives spread into low-SNR, implausibly-large territory — the fingerprint of eclipsing binaries.

Leakage audit (the most important cleaning step). A dataset like this is full of columns that secretly encode the answer. The organisers had already removed `koi_score`. I removed several more: `kepler_name` (a giveaway that a planet is confirmed), `koi_pdisposition` (the Kepler pipeline's *own* verdict on the same question), and vetting metadata recorded *after* the disposition was decided. The four Robovetter false-positive flags are legitimate diagnostics but heavily encode 'is this a false positive,' so I treated them as a toggleable group and *measured* their effect instead of letting them quietly dominate. This discipline is what separates a model that merely scores well from one a scientist can trust.

3 Feature Engineering

I built domain-aware features describing whether a transit is physically consistent with a real planet. (a) **Fractional measurement uncertainties** (error ÷ value) recycle the ~130 discarded error columns into signal-quality proxies — a poorly-measured signal is a suspicious one. (b) **Log transforms** tame quantities that span many orders of magnitude (period, depth, insolation, radius, SNR). (c) **Physics-consistency ratios**, most notably `depth_consistency` (observed transit depth versus the depth implied by the fitted planet-to-star size), which flags the tell-tale mismatch of a blended eclipsing binary; plus `duty_cycle`, `snr_per_transit`, and `mes_ses_ratio`. SHAP later confirmed several of these engineered features rank among the model's top predictors.

4 Modelling & Performance

I compared four models under an identical stratified 80/20 split — Logistic Regression, Random Forest, HistGradientBoosting, and XGBoost — then tuned XGBoost with a 40-configuration randomized search (which confirmed my configuration was already near-optimal) and validated it with 5-fold cross-validation. XGBoost was chosen as the final model: it handles missing values, captures non-linear feature interactions, is robust to the collinearity in these physical measurements, and gave the strongest, most stable score. The tight ±0.007 spread across folds shows the result generalises rather than fitting one lucky split.

Class	Precision	Recall	F1-score	Support
FALSE POSITIVE	0.991	0.996	0.993	968
CONFIRMED	0.916	0.913	0.914	549
CANDIDATE	0.873	0.866	0.869	396
Accuracy			0.945	1913
Macro avg	0.926	0.925	0.926	1913
Weighted avg	0.945	0.945	0.945	1913

Table 1 — Held-out test-set performance of the tuned XGBoost (with Robovetter flags).

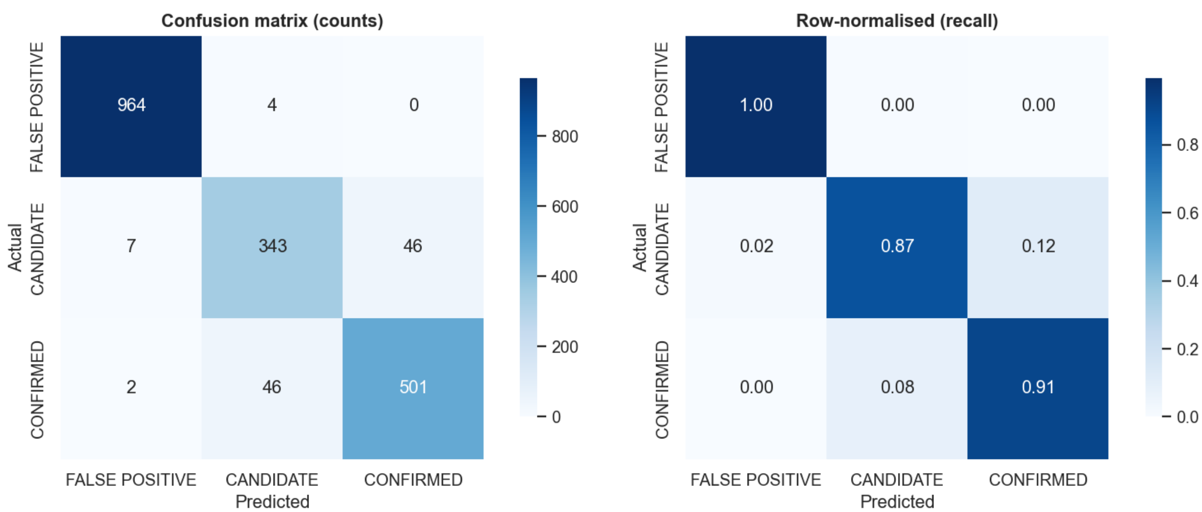

Figure 1 — Confusion matrix. FALSE POSITIVE is recovered almost perfectly (~99% recall); virtually all remaining error is CANDIDATE ↔ CONFIRMED confusion.

5 Explainability

The model is not a black box. I interpret it three ways: XGBoost gain importance, model-agnostic permutation importance (the drop in macro-F1 when a feature is shuffled), and SHAP — globally, per class, and for individual predictions. **In plain language, the model asks three questions of every dip in starlight:** (1) *Does it even look like a planet transit?* — signal-to-noise and the Robovetter flags throw out weak or non-transit-shaped signals. (2) *Is the geometry believable?* — transit depth, planet radius, and `depth_consistency` catch eclipsing binaries whose 'planet' would be absurdly large. (3) *Is the signal strong and repeated enough to trust?* — number of transits and SNR separate solid detections from marginal ones. The chart below reads out exactly which measurements drove one real verdict.

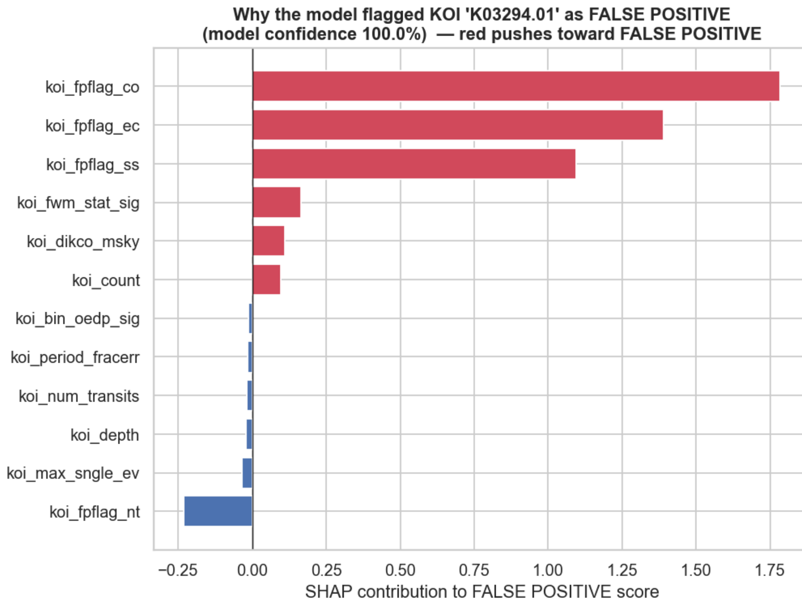

Figure 2 — A single-prediction SHAP explanation: off-centre starlight (centroid-offset and ephemeris-match flags) exposed a background eclipsing binary. False positive, explained.

6 Key Findings & Honest Limits

The confusion matrix tells a clean story, and my with/without-flags experiment explains it: the diagnostic flags make false-positive rejection almost trivial, but they — and every other photometric feature — are nearly useless for the CONFIRMED-versus-CANDIDATE split. That is because the only real difference between a candidate and a confirmed planet is whether astronomers have *finished the follow-up work*, which the light curve simply does not encode. Reframed as the binary ‘real signal versus false positive’ question, the same model reaches macro-F1 0.993. So impostor rejection is nearly solved science, while confirmation status is an irreducible uncertainty from photometry alone — and rather than hide this behind a single number, the model surfaces it.

7 Reproducibility

Everything is seeded (`random_state=42`) and version-pinned in `requirements.txt`. The single self-contained notebook regenerates every number and figure top-to-bottom on a fresh machine; the public GitHub repository holds the notebook, this report, and all plots.